

# Yair Schiff

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[yair-schiff.github.io](https://yair-schiff.github.io)

## RESEARCH INTERESTS

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Generative modeling, Optimal transport, AI for Science, AI for Social Good



## EDUCATION

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**CORNELL UNIVERSITY** – DEPARTMENT OF COMPUTER SCIENCE, New York, NY

2022 – Present

*PhD candidate in Computer Science*

*Advisor: Professor Volodymyr Kuleshov*

- *Fellowships:* Hal & Inge Marcus PhD Fellowship (2022-2023)
- *Awards:* Cornell Tech Outstanding TA Award (2023)
- *Teaching:* Applied Machine Learning (TA Fall 2022), Deep Learning (TA Spring 2023)

**NEW YORK UNIVERSITY** – COURANT INSTITUTE OF MATHEMATICAL SCIENCES, New York, NY

May 2019

*MS in Computer Science, GPA: 3.97/4.00*

- *Relevant Coursework:* Advanced Machine Learning, Artificial Intelligence, Computer Vision, Deep Learning, Graphical Processing Units, Machine Learning, Mathematics of Deep Learning, Predictive Analytics

**UNIVERSITY OF PENNSYLVANIA** – COLLEGE OF ARTS AND SCIENCES, Philadelphia, PA

May 2014

*BA Summa Cum Laude with Distinction in Economics, GPA: 3.93/4.00*

- *Academic Honors:* Phi Beta Kappa, Dean's list 2010-2014



## RESEARCH AND PUBLICATIONS

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### PUBLICATIONS

d2: Improved Techniques for Training Reasoning Diffusion Language Models

[ICML 2026](#)

Guanghan Wang, Gilad Turok, **Yair Schiff**, Volodymyr Kuleshov

Encoder-Decoder Diffusion Language Models for Efficient Training and Inference

[NeurIPS 2025](#)

Marianne Arriola\*, **Yair Schiff**\*, Hao Phung, Aaron Gokaslan, Volodymyr Kuleshov

(\*equal contribution)

Remasking Discrete Diffusion Models with Inference-Time Scaling

[NeurIPS 2025](#)

Guanghan Wang\*, **Yair Schiff**\*, Subham Sekhar Sahoo, Volodymyr Kuleshov

(\*equal contribution)

Cross-species plant genomes modeling at single nucleotide resolution

[PNAS](#)

using a pre-trained DNA language model

Jingjing Zhai, Aaron Gokaslan, **Yair Schiff**, Ana Berthel, Zong-Yan Liu, Zachary R Miller

Armin Scheben, Michelle C Stitzer, Cinta Romay, Edward S. Buckler, Volodymyr Kuleshov

Simple Guidance Mechanisms for Discrete Diffusion Models

[ICLR 2025](#)

**Yair Schiff**, Subham Sekhar Sahoo, Guanghan Wang, Hao Phung,

Sam Boshar, Hugo Dalla-torre, Bernardo P de Almeida,

Alexander Rush, Thomas Pierrot, Volodymyr Kuleshov

Simple and Effective Masked Diffusion Language Models

[NeurIPS 2024](#)

Subham Sekhar Sahoo, Marianne Arriola, **Yair Schiff**, Aaron Gokaslan,

Edgar Marroquin, Justin T Chiu, Alexander Rush, Volodymyr Kuleshov

Auditing and Generating Synthetic Data with Controllable Trust Trade-offs

[IEEE JETCAS](#)

Brian Belgodere, Pierre Dognin, Adam Ivankay, Igor Melnyk, Youssef Mroueh, Aleksandra Mojsilovic, Jiri Navartil, Apoorva Nitsure, Inkit Padhi, Mattia Rigotti, Jarret Ross, **Yair Schiff**, Radhika Vedpathak, Richard A. Young

Caduceus: Bi-Directional Equivariant Long-Range  
DNA Sequence Modeling

[ICML 2024](#)

**Yair Schiff**, Chia-Hsiang Kao, Aaron Gokaslan, Tri Dao, Albert Gu, Volodymyr Kuleshov

DySLIM: Dynamics Stable Learning by Invariant Measure for Chaotic Systems

[ICML 2024](#)

**Yair Schiff**, Zhong Yi Wan, Jeffrey B. Parker, Stephan Hoyer, Volodymyr Kuleshov, Fei Sha, Leonardo Zepeda-Núñez

InfoDiffusion: Representation Learning

[ICML 2023](#)

Using Information Maximizing Diffusion Models

Yingheng Wang, **Yair Schiff**, Aaron Gokaslan, Weishen Pan, Fei Wang, Christopher De Sa, Volodymyr Kuleshov

Semi-Autoregressive Energy Flows: Exploring Likelihood-Free  
Training of Normalizing Flows

[ICML 2023](#)

Phillip Si, Zeyi Chen, Subham Sekhar Sahoo, **Yair Schiff**, Volodymyr Kuleshov

Learning with Stochastic Orders

[ICLR 2023](#)

Carles Domingo-Enrich, **Yair Schiff**, Youssef Mroueh

(Notable Top 25% acceptance)

Semi-Parametric Inducing Point Networks and Neural Processes

[ICLR 2023](#)

Richa Rastogi, **Yair Schiff**, Alon Hachohen, Zhaozhi Li, Ian Lee, Yuntian Deng, Mert R. Sabuncu, Volodymyr Kuleshov

Cloud-Based Real-Time Molecular Screening Platform  
with MolFormer

[ECML PKDD 2022 Demo Track](#)

Brian Belgodere\*, Vijil Chenthamarakshan\*, Payel Das\*, Pierre Dognin\*, Toby Kurien\*, Igor Melnyk\*, Youssef Mroueh\*, Inkit Padhi\*, Mattia Rigotti\*, Jarret Ross\*, **Yair Schiff**\*, Richard A. Young\*

(\*alphabetical order  
equal contribution)

Optimizing Functionals on the Space of Probabilities with  
Input Convex Neural Networks

[Transactions of Machine Learning Research](#)

David Alvarez-Melis, **Yair Schiff**, Youssef Mroueh

Augmenting Molecular Deep Generative Models with  
Topological Data Analysis Representations

[ICASSP 2022](#)

**Yair Schiff**\*, Vijil Chenthamarakshan\*, Samuel Hoffman\*, Karthikeyan Natesan Ramamurthy\*, Payel Das\*

(\*equal contribution)

Predicting Deep Neural Network Generalization with Perturbation Response Curves

[NeurIPS 2021](#)

**Yair Schiff**, Brian Quanz, Payel Das, Pin-Yu Chen

Tabular Transformers for Modeling Multivariate Time Series

[ICASSP 2021](#)

Inkit Padhi, **Yair Schiff**, Igor Melnyk, Mattia Rigotti, Youssef Mroueh, Pierre Dognin, Jarret Ross, Ravi Nair, Erik Altman

Image Captioning as an Assistive Technology: Lessons Learned from  
VizWiz 2020 Challenge

[Journal of AI Research](#)

Pierre Dognin\*, Igor Melnyk\*, Youssef Mroueh\*, Inkit Padhi\*, Mattia Rigotti\*, Jarret Ross\*, **Yair Schiff**\*, Richard Young, Brian Belgodere

(\*alphabetical order  
equal contribution)

## WORKSHOPS

Advancing DNA Language Models: The Genomics Long-Range Benchmark  
*Evan Trop, Chia-Hsiang Kao, Mckinley Polen, **Yair Schiff**, Bernardo P. de Almeida, Aaron Gokaslan, Thomas Pierrot, Volodymyr Kuleshov*

[AAAI Workshop 2024](#)  
[ICLR Workshop 2024](#)

Optimizing Functionals on the Space of Probabilities with  
Input Convex Neural Networks  
*David Alvarez-Melis, **Yair Schiff**, Youssef Mroueh*

[NeurIPS Workshop 2021](#)  
*Spotlight presentation*

Gi and Pal Scores: Deep Neural Network Generalization Statistics  
***Yair Schiff**, Brian Quanz, Payel Das, Pin-Yu Chen*

[ICLR Workshop 2021](#)

Characterizing the Latent Space of Molecular Deep Generative Models  
with Persistent Homology Metrics  
***Yair Schiff**, Vijil Chenthamarakshan, Karthikeyan Natesan Ramamurthy, Payel Das*

[NeurIPS Workshop 2020](#)  
*Spotlight presentation*

Alleviating Noisy Data in Image Captioning with Cooperative Distillation  
*Pierre Dognin\*, Igor Melnyk\*, Youssef Mroueh\*, Inkit Padhi\*, Mattia Rigotti\*, Jarret Ross\*, **Yair Schiff**\**

[CVPR Workshop 2020](#)  
*(\*alphabetical order,  
equal contribution)*

## TALKS AND PRESENTATIONS

Topological Data Analysis and Beyond Workshop

[NeurIPS 2020](#)

- Presented spotlight poster “Characterizing the Latent Space of Molecular Deep Generative Models” ([video](#))

VizWiz Grand Challenge Workshop

[CVPR 2020](#)

- Presented winning submission to VizWiz Grand Challenge ([video](#))
- Presented “Alleviating Noisy Data in Image Captioning with Cooperative Distillation” ([video](#))

## OPEN-SOURCE CONTRIBUTIONS

Caduceus

[Github.com](#)

- Authored codebase for long-range DNA sequence modeling using newly proposed Caduceus model.

swirl-dynamics

[Github.com](#)

- Added a project for modeling dynamical systems using a regularized objective that aims to preserve systems’ invariant measures

SPIN: Semi-Parametric Inducing Point Networks and Neural Process

[Github.com](#)

- Implemented Inducing Point Neural Processes and wrote code for training and evaluation

TabFormer: Tabular Transformers for Modeling Multivariate Time Series

[Github.com](#)

- Wrote code for training and evaluating GPT-like models on tabular data to generate new, synthetic data that matches the underlying distributions of the real table variables

pytorch-PPUU: Prediction and Policy-learning Under Uncertainty

[Github.com](#)

- Added a new dataset on which the self-driving policy could be trained
- Enhanced the self-driving vehicle’s policy to enable dynamic lane changes

## PROFESSIONAL SERVICES

- ICML 2026 reviewer – Gold Reviewer
- NeurIPS 2025 reviewer – Top Reviewer
- ICML 2025 reviewer

- ICLR 2025 reviewer
- ICML 2024 reviewer
- NeurIPS 2023 reviewer – Top Reviewer



## WORK EXPERIENCE

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**INSTADEEP**, New York, NY

May 2024 – Nov 2025

*PhD Researcher Intern*

- Applying generative modeling techniques to genomic sequences
- Investigating control mechanisms for guided sequence generation

**GOOGLE**, New York, NY

May 2023 – Oct 2023

*Student Researcher*

- Researched new methods for stabilizing autoregressive rollouts of dynamical system models
- Contributed to internal Google and open-source libraries for modeling dynamical systems

**IBM WATSON MACHINE LEARNING**, New York, NY

Aug 2019 – Aug 2022

*Cognitive Software Developer*

- Contributed to continuous development and testing of Watson Machine Learning products
- Facilitated weekly Cloud releases, Cloud Pak for Data platform releases, and the launch of AutoAI feature engineering on relational data, AutoAI Time Series, AutoAI Notebooks, and Federated Learning products
- Published 6 medium.com articles about Watson Machine Learning product releases
- Received Outstanding Technical Achievement Award for work on the release of AutoAI feature engineering on relational data
- Received CrushIT Team Excellence Award as part of the Watson Machine Learning Training team

*Research Contributor to IBM Research AI Challenges*

Aug 2019 – Aug 2022

- Volunteered to contribute to IBM Research AI challenges, working with the Trusted AI Department
- Member of the first-place winning team in the 2020 VizWiz Grand Challenge: Image Captioning as an Assistive Technology for the Visually Impaired
- Co-authored with the Trusted AI team on several publications in the fields of Generative Modeling, Molecular Discovery, Deep Learning Generalization, and AI for Social Good
- Received two 2021 IBM Research Accomplishments awards for contributions to (1) trustworthy AI generative modeling and (2) deployment of large-scale transformer models on OpenShift environments

**SIMON-KUCHER AND PARTNERS**, New York, NY

Sept 2014 – Aug 2017

*Consultant*

- Advised global companies spanning various industries – including internet, media, consumer electronic goods, and chemicals – on areas for better revenue capture
- Synthesized large data sets (e.g., 300 million+ client transactions), customer research (surveys & conjoint studies sent to thousands of respondents), and secondary research to create solutions to client needs



## GROUPS AND AFFILIATIONS

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**GRADS FOR GENDER INCLUSION IN COMPUTING**, New York, NY

Present

*Member*

- Member of a group dedicated to combating harassment, pushing for policy change, and creating supportive spaces

**STUDENT-APPLICANT SUPPORT PROGRAM**, New York, NY

Present

*Volunteer*

- Provide feedback and support to students applying to graduate programs in Computer Science and related fields

**PHD PEER MENTOR PROGRAM**, New York, NY

Present

*Mentor*

- Meet with fellow PhD student to provide guidance about graduate school life at Cornell Tech



## **SKILLS**

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- *Programming Languages:* C++, Java, Python
- *Deep Learning Frameworks:* PyTorch, PyTorch Lightning
- *Data Tools:* Excel, Stata, Tableau
- *Foreign Languages:* Fluent in Hebrew